Theoretical Minimum General Relativity Exercises

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Abstract

These are some exercises arising from the *General Relativity*[1] lectures from Leonard Susskind's *Theoretical Minimum* series[2]. Section 1 contains a derivation of the Schwarzschild metric. Section 2 examines linearized solutions and gravitational waves.

Disclaimer: I have created these notes as an aide-mémoire for my own use; if you find them useful, you are welcome, but I'd appreciate hearing from you. They are not intended as a substitute for listening to the lectures. The intellectual property for all material derived from the lectures belongs, of course, to Professor Susskind; any mistakes, however, are my own.

The notes were created using TexStudio[3], which I recommend for compiling them to a PDF, and the bibliography was created using JabRef[4].

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1 Schwarzschild Metric

The Schwarzschild Metric is presented, without proof, in [1, Lecture 6]. In this section the coordinates are denoted: (t, r, θ, ϕ). The summation convention is observed for all other indices: e.g. ${\Gamma^\alpha}_{\gamma\alpha}$ denoted summation over the dummy index α, but ${\Gamma^\theta}_{\gamma\theta}$ is not summed.

1.1 Ansatz

We will seek a solution of the form[5]:

$$\begin{aligned} dt^2 =& e^{\nu(r)}dt^2 - e^{\lambda(r)}dr^2 - r^2 d\Omega^2, \text{ where} \\ d\Omega^2 =& d\theta^2 + cos^2\theta d\phi^2 \end{aligned} \tag{1}$$

The non-zero $g_{\mu\nu}$ are:

$$\begin{aligned} g_{tt} =& e^{\nu(r)} \\ g_{rr} =& -e^{\lambda(r)} \\ g_{\theta\theta} =& -r^2 \\ g_{\phi\phi} =& -r^2\cos^2\theta \end{aligned}$$

The $g^{\mu\nu}$ are the inverse of the $g_{\mu\nu}$, so:

$$\begin{aligned} g^{tt} =& e^{-\nu(r)} \\ g^{rr} =& -e^{-\lambda(r)} \\ g^{\theta\theta} =& -\frac{1}{r^2} \\ g^{\phi\phi} =& -\frac{1}{r^2\cos^2\theta} \end{aligned}$$

The only non-zero derivatives of $g^{\mu\nu}$ are:

$$\begin{aligned}
\partial_r g_{tt} =& \nu'(r) e^{\nu(r)} \\
\partial_r g_{rr} =& -\lambda'(r) e^{\lambda(r)} \\
\partial_r g_{\theta\theta} =& -2r \\
\partial_r g_{\phi\phi} =& -2r\cos^2\theta \\
\partial_\theta g_{\phi\phi} =& 2r^2\cos\theta\sin\theta
\end{aligned}$$

1.2 Christoffel Symbols

We start be enquiring which Christoffel are not identically zero. For each ρ we ask whether there are any (μ, ν) for which the terms making up $\Gamma^\rho{}_{\mu\nu}$ are non-zero.

$$\begin{aligned}
\Gamma^t{}_{\mu\nu} =& \frac{1}{2}\Big[\underbrace{\partial_\mu g_{t\nu}}_{\mu=r,\,\nu=t} + \underbrace{\partial_\nu g_{\mu t}}_{\mu=t,\,\nu=r} - \underbrace{\partial_t g_{\mu\nu}}_{\equiv 0} \Big] e^{-\nu(r)} \\
\Gamma^t{}_{rt} =& \frac{1}{2}\partial_r g_{tt} e^{-\nu(r)} \\
=& \frac{1}{2}\nu'(r) \cancel{e^{\nu(r)}} \cancel{e^{-\nu(r)}} \\
=& \frac{\nu'(r)}{2} \\
\Gamma^t{}_{tr} =& \frac{\nu'(r)}{2} \\
\Gamma^t{}_{rt} =& \Gamma^t{}_{tr} = \frac{\nu'(r)}{2}
\end{aligned} \tag{2}$$

$$
\begin{aligned}
\Gamma^r{}_{\mu\nu} &= -\frac{1}{2}\big[\underbrace{\partial_\mu g_{r\nu}}_{\mu=\nu=r} + \underbrace{\partial_\nu g_{\mu r}}_{\mu=\nu=r} - \underbrace{\partial_r g_{\mu\nu}}_{\mu=\nu}\big]e^{-\lambda(r)} \\
\Gamma^r{}_{tt} &= -\frac{1}{2}\big[-\partial_r g_{tt}\big]e^{-\lambda(r)} \\
&= \frac{1}{2}\big[\nu'(r)e^{\nu(r)}\big]e^{-\lambda(r)} \\
&= \frac{1}{2}\nu'(r)e^{\nu(r)-\lambda(r)} \qquad (3) \\
\Gamma^r{}_{rr} &= -\frac{1}{2}\big[\partial_r g_{rr}\big]e^{-\lambda(r)} \\
&= \frac{1}{2}\big[\lambda'(r)\cancel{e^{\lambda(r)}}\big]\cancel{e^{-\lambda(r)}} \\
&= \frac{\lambda'(r)}{2} \qquad (4) \\
\Gamma^r{}_{\theta\theta} &= -\frac{1}{2}\big[-\partial_r g_{\theta\theta}\big]e^{-\lambda(r)} \\
&= \frac{1}{\cancel{2}}\big[-\cancel{2}r\big]e^{-\lambda(r)} \\
&= -re^{-\lambda(r)} \qquad (5) \\
\Gamma^r{}_{\phi\phi} &= -\frac{1}{2}\big[-\partial_r g_{\phi\phi}\big]e^{-\lambda(r)} \\
&= \frac{1}{\cancel{2}}\big[-\cancel{2}r\cos^2\theta\big]e^{-\lambda(r)} \\
&= -r\cos^2\theta e^{-\lambda(r)} \qquad (6)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma^\theta{}_{\mu\nu} &= -\frac{1}{2}\big[\underbrace{\partial_\mu g_{\theta\nu}}_{\mu=r,\,\nu=\theta} + \underbrace{\partial_\nu g_{\mu\theta}}_{\mu=\theta,\,\nu=r} - \underbrace{\partial_\theta g_{\mu\nu}}_{\mu=\nu=\phi}\big]\frac{1}{r^2} \\
\Gamma^\theta{}_{r\theta} &= -\frac{1}{2}\big[\partial_r g_{\theta\theta}\big]\frac{1}{r^2} \\
&= -\frac{1}{2}\big[-2r\big]\frac{1}{r^2} \\
&= \frac{1}{r} \\
\Gamma^\theta{}_{r\theta} &= \Gamma^\theta{}_{\theta r} = \frac{1}{r} \qquad (7) \\
\Gamma^\theta{}_{\phi\phi} &= -\frac{1}{2}\big[-\partial_\theta g_{\phi\phi}\big]\frac{1}{r^2} \\
&= \frac{1}{\cancel{2}}\big[\cancel{2}\cancel{r^2}\cos\theta\sin\theta\big]\frac{1}{\cancel{r^2}} \\
&= \cos\theta\sin\theta \qquad (8)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma^{\phi}{}_{\mu\nu} =& -\frac{1}{2}\Big[\underbrace{\partial_\mu g_{\phi\nu}}_{\mu\in\{\theta,r\},\,\nu=\phi} + \underbrace{\partial_\nu g_{\mu\phi}}_{\mu=\phi,\,\nu\in\{\theta,r\}} - \underbrace{\partial_\phi g_{\mu\nu}}_{\equiv 0}\Big]\frac{1}{r^2\cos^2\theta} \\
\Gamma^{\phi}{}_{\theta\phi} =& -\frac{1}{2}\big[\partial_\theta g_{\phi\phi}\big]\frac{1}{r^2\cos^2\theta} \\
=& -\frac{1}{2}\big[2r^2\cos\theta\sin\theta\big]\frac{1}{r^2\cos^2\theta} \\
=& -\tan\theta \\
\Gamma^{\phi}{}_{\theta\phi} =& \Gamma^{\phi}{}_{\phi\theta} = -\tan\theta \qquad (9) \\
\Gamma^{\phi}{}_{r\phi} =& -\frac{1}{2}\big[\partial_r g_{\phi\phi}\big]\frac{1}{r^2\cos^2\theta} \\
=& \cancel{-}\frac{1}{\cancel{2}}\big[\cancel{-}\cancel{2}r\cancel{\cos^2\theta}\big]\frac{1}{r^2\cancel{\cos^2\theta}} \\
=& \frac{1}{r} \\
\Gamma^{\phi}{}_{r\phi} =& \Gamma^{\phi}{}_{\phi r} = \frac{1}{r} \qquad (10)
\end{aligned}
$$

1.3 Riemann & Ricci Tensors

From [6, Lecture II]

$$
\begin{aligned}
R^{\mu}{}_{\nu\alpha\beta} =& \partial_\alpha\Gamma^{\mu}{}_{\nu\beta} - \partial_\beta\Gamma^{\mu}{}_{\nu\alpha} + \Gamma^{\mu}{}_{\gamma\alpha}\Gamma^{\gamma}{}_{\nu\beta} - \Gamma^{\mu}{}_{\gamma\beta}\Gamma^{\gamma}{}_{\nu\alpha} \\
R_{\mu\nu} =& R^{\alpha}{}_{\mu\alpha\nu} \\
=& \partial_\alpha\Gamma^{\alpha}{}_{\mu\nu} - \partial_\nu\Gamma^{\alpha}{}_{\mu\alpha} + \Gamma^{\alpha}{}_{\gamma\alpha}\Gamma^{\gamma}{}_{\mu\nu} - \Gamma^{\alpha}{}_{\gamma\nu}\Gamma^{\gamma}{}_{\mu\alpha} \\
R =& R_{\mu\nu}g^{\mu\nu}
\end{aligned}
$$

We compute the components of $R_{\mu\nu}$.

1.3.1 R_{tt} and R_{rr}

Since (1) contains two unknown functions, I expect to be able to determine them from two components of $R_{\mu\nu}$.

$$
\begin{aligned}
R_{tt} =&\partial_\alpha \Gamma^\alpha{}_{tt} - \underbrace{\partial_t \Gamma^\alpha{}_{t\alpha}}_{=0} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{tt} - \Gamma^\alpha{}_{\gamma t}\Gamma^\gamma{}_{t\alpha} \\
=&\partial_r \Gamma^r{}_{tt} + \big[\Gamma^t{}_{rt} + \Gamma^r{}_{rr} + \Gamma^\theta{}_{r\theta} + \Gamma^\phi{}_{r\phi}\big]\Gamma^r{}_{tt} - 2\Gamma^t{}_{rt}\Gamma^r{}_{tt} \\
=&\partial_r \Gamma^r{}_{tt} + \big[-\Gamma^t{}_{rt} + \Gamma^r{}_{rr} + \Gamma^\theta{}_{r\theta} + \Gamma^\phi{}_{r\phi}\big]\Gamma^r{}_{tt} \\
=&\partial_r \big[\frac{1}{2}\nu'(r)e^{\nu(r)-\lambda(r)}\big] + \big[-\frac{\nu'(r)}{2} + \frac{\lambda'(r)}{2} + \frac{1}{r} + \frac{1}{r}\big]\frac{1}{2}\nu'(r)e^{\nu(r)-\lambda(r)} \\
=&\frac{1}{2}\big[\nu''(r) + \nu'(r)\big(\nu'(r) - \lambda'(r)\big)\big]e^{\nu(r)-\lambda(r)} + \big[-\frac{\nu'(r) - \lambda'(r)}{2} + \frac{2}{r}\big]\frac{1}{2}\nu'(r)e^{\nu(r)-\lambda(r)} \\
=&\frac{1}{2}\left[\nu''(r) + \frac{1}{2}\nu'(r)\big(\nu'(r) - \lambda'(r)\big) + \frac{2}{r}\nu'(r)\right]e^{\nu(r)-\lambda(r)} \qquad (11)
\end{aligned}
$$

$$
\begin{aligned}
R_{rr} =&\partial_\alpha \Gamma^\alpha{}_{rr} - \partial_r \Gamma^\alpha{}_{r\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{rr} - \Gamma^\alpha{}_{\gamma r}\Gamma^\gamma{}_{r\alpha} \\
=&\underbrace{\partial_t \Gamma^t{}_{rr}}_{=0} + \cancel{\partial_r \Gamma^r{}_{rr}} + \underbrace{\partial_\theta \Gamma^\theta{}_{rr}}_{=0} + \underbrace{\partial_\phi \Gamma^\phi{}_{rr}}_{=0} \\
&- \partial_r \Gamma^t{}_{rt} - \cancel{\partial_r \Gamma^r{}_{rr}} - \partial_r \Gamma^\theta{}_{r\theta} - \partial_r \Gamma^\phi{}_{r\phi} \\
&+ \big[\Gamma^t{}_{rt} + \cancel{\Gamma^r{}_{rr}} + \Gamma^\theta{}_{r\theta} + \Gamma^\phi{}_{r\phi}\big]\Gamma^r{}_{rr} \\
&- \Gamma^t{}_{tr}\Gamma^t{}_{rt} - \cancel{\Gamma^r{}_{rr}\Gamma^r{}_{rr}} - \Gamma^\theta{}_{\theta r}\Gamma^\theta{}_{r\theta} - \Gamma^\phi{}_{\phi r}\Gamma^\phi{}_{r\phi} \\
=&-\partial_r \frac{\nu'(r)}{2} - \partial_r \frac{1}{r} - \partial_r \frac{1}{r} + \big[\frac{\nu'(r)}{2} + \frac{1}{r} + \frac{1}{r}\big]\frac{\lambda'(r)}{2} - \big(\frac{\nu'(r)}{2}\big)^2 - \frac{1}{r^2} - \frac{1}{r^2} \\
=&-\frac{\nu''(r)}{2} + \cancel{\frac{2}{r^2}} + \left(\frac{\nu'(r)}{2} + \frac{2}{r}\right)\frac{\lambda'(r)}{2} - \big(\frac{\nu'(r)}{2}\big)^2 - \cancel{\frac{2}{r^2}} \\
=&-\frac{1}{2}\left[\nu''(r) - \left(\frac{\nu'(r)}{2} + \frac{2}{r}\right)\lambda'(r) + \frac{\big(\nu'(r)\big)^2}{2}\right] \qquad (12)
\end{aligned}
$$

1.3.2 Determination of ν and λ

We now have enough information to determine ν and λ. From (11) and (12)

$$
\begin{aligned}
\nu''(r) + \frac{1}{2}\nu'(r)\big(\nu'(r) - \lambda'(r)\big) + \frac{2}{r}\nu'(r) =&0 \qquad (13) \\
\nu''(r) - \left(\frac{\nu'(r)}{2} + \frac{2}{r}\right)\lambda'(r) + \frac{\big(\nu'(r)\big)^2}{2} =&0
\end{aligned}
$$

Subtracting these equations:

$$
\cancel{\nu''(r)} + \frac{1}{2}\nu'(r)\big(\nu'(r) - \lambda'(r)\big) + \frac{2}{r}\nu'(r) - \cancel{\nu''(r)} + \left(\frac{\nu'(r)}{2} + \frac{2}{r}\right)\lambda'(r) - \frac{\big(\nu'(r)\big)^2}{2} =0
$$

$$
\begin{aligned}
\frac{1}{2}\nu'(r)\big(\cancel{\nu'(r)} - \cancel{\lambda'(r)}\big) + \frac{2}{r}\nu'(r) + \left(\cancel{\frac{\nu'(r)}{2}} + \frac{2}{r}\right)\lambda'(r) - \cancel{\frac{\big(\nu'(r)\big)^2}{2}} =& 0 \\
\frac{2}{r}\big(\nu'(r) + \lambda'(r)\big) =& 0 \\
\nu'(r) + \lambda'(r) =& 0, \text{ so for some constant } c \\
e^{\lambda}(r) =& \frac{c}{e^{\nu}(r)} \qquad (14)
\end{aligned}
$$

Substituting in (13):

$$
\begin{aligned}
\nu''(r) + \nu'(r)^2 + \frac{2}{r}\nu'(r) =& 0. \text{ Now} \qquad (15) \\
\big[re^{\nu(r)}\big]'' =& \big[e^{\nu} + r\nu' e^{\nu}\big]' \\
=& \big[e^{\nu}\nu' + e^{\nu}\nu' + r\nu'' e^{\nu} + r(\nu')^2 e^{\nu}\big] \\
=& re^{\nu}\big[\frac{2}{r} + \nu'' + (\nu')^2\big], \text{ so (15) becomes:} \\
\big[re^{\nu(r)}\big]'' =& 0, , \text{ or} \\
re^{\nu(r)} =& ar + b, \text{ for some constants } a \text{ and } b \\
e^{\nu(r)} =& a + \frac{b}{r}
\end{aligned}
$$

Now we know that the metric (1) should be Minkowskian at infinity, which requires $\nu(r) \to 1$ as $r \to \infty$, so from:

$$
\begin{aligned}
dt^2 =& \big(a + \frac{b}{r}\big)dt^2 - \frac{c}{a + \frac{b}{r}}dr^2 - r^2 d\Omega^2, \text{ for some we see that:} \\
a =& \frac{c}{a} = 1, \text{ using (14), i.e.} \\
c =& 1
\end{aligned}
$$

We saw in [1, Lecture 5] that we need $g_{00} \approxeq 1 - \frac{2MG}{r}$ for small r, in order to get agreement with Newtonian gravity, hence $b = -2MG$, and (1) becomes:

$$
dt^2 = \big(1 - \frac{2MG}{R}\big)dt^2 - \frac{dr^2}{1 - \frac{2MG}{R}} - r^2 d\Omega^2 \qquad (16)
$$

1.3.3 $R_{\theta\theta}$ and $R_{\phi\phi}$

We need to establish that the remaining diagonal elements are zero using the metric (16). We will find the following Lemma useful:

Lemma 1. $\partial_r\big[re^{-\lambda(r)}\big] = 1$

Proof. From (16)

$$
\begin{aligned}
e^{\lambda(r)} =& \frac{1}{1-\frac{2MG}{r}}, \text{ whence} \\
\partial_r\big[re^{-\lambda(r)}\big] =& \partial_r\big[r(1-\frac{2MG}{r})\big] \\
=& \partial_r[r-2MG] \\
=& 1
\end{aligned}
$$

□

$$
\begin{aligned}
R_{\theta\theta} =& \partial_\alpha \Gamma^\alpha{}_{\theta\theta} - \partial_\theta \Gamma^\alpha{}_{\theta\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{\theta\theta} - \Gamma^\alpha{}_{\gamma\theta}\Gamma^\gamma{}_{\theta\alpha} \\
=& \partial_r \Gamma^r{}_{\theta\theta} - \partial_\theta\big[\Gamma^\phi{}_{\theta\phi} + \Gamma^r{}_{\theta r}\big] + \Big(\Gamma^t{}_{rt} + \Gamma^r{}_{rr} + \cancel{\Gamma^\theta{}_{r\theta}} + \Gamma^\phi{}_{r\phi}\Big)\Gamma^r{}_{\theta\theta} \\
& - \cancel{\Gamma^r{}_{\theta\theta}\Gamma^\theta{}_{\theta r}} - \Gamma^\theta{}_{r\theta}\Gamma^r{}_{\theta\theta} - \Gamma^r{}_{r\theta}\Gamma^r{}_{\theta r} - \Gamma^\phi{}_{\phi\theta}\Gamma^\phi{}_{\phi\theta} \\
=& \partial_r\Big[-re^{-\lambda(r)}\Big] - \partial_\theta\Big[-\tan\theta + \frac{1}{r}\Big] \\
& + \Big(\underbrace{\frac{\nu'(r)}{2} + \frac{\lambda'(r)}{2}}_{=0 \text{ from } (14)} + \cancel{\frac{1}{r}}\Big)\cancel{\Big[-re^{-\lambda(r)}\Big]} \\
& + \cancel{\frac{re^{-\lambda(r)}}{r}} - \frac{1}{r^2} - \tan^2\theta \qquad (17) \\
& - \underbrace{1}_{\text{From Lemma 1}} + \underbrace{\sec^2\theta}_{=1+\tan^2\theta} + \cancel{\frac{1}{r^2}} - \cancel{\frac{1}{r^2}} - \tan^2\theta \\
=& 0, \text{ as expected.}
\end{aligned}
$$

Since the calculation of $R_{\phi\phi}$ has proved complex, I have expanded the terms, and used the cancel symbol to mark vanishing Christoffel symbols.

$$
\begin{aligned}
\partial_\alpha \Gamma^\alpha{}_{\phi\phi} - \partial_\phi \Gamma^\alpha{}_{\phi\alpha} =& \partial_r\big[-\cos^2\theta re^{-\lambda(r)}\big] + \partial_\theta\big[\cos\theta\sin\theta\big] - \underbrace{\partial_\phi \Gamma^\alpha{}_{\phi\alpha}}_{=0} \\
=& -\cos^2\theta \cdot \underbrace{\partial_r\big[re^{-\lambda(r)}\big]}_{=1 \text{ from Lemma 1}} + \partial_\theta\big[\cos\theta\sin\theta\big] \\
=& -\cancel{\cos^2\theta} - \sin^2\theta + \cancel{\cos^2\theta} \qquad (18)
\end{aligned}
$$

$$
\begin{aligned}
\Gamma^{\alpha}{}_{\gamma\alpha}\Gamma^{\gamma}{}_{\phi\phi} =& \Gamma^{\alpha}{}_{t\alpha}\cancel{\Gamma^{t}{}_{\phi\phi}} + \Gamma^{\alpha}{}_{r\alpha}\Gamma^{r}{}_{\phi\phi} + \Gamma^{\alpha}{}_{\theta\alpha}\Gamma^{\theta}{}_{\phi\phi} + \Gamma^{\alpha}{}_{\phi\alpha}\cancel{\Gamma^{\phi}{}_{\phi\phi}} \\
=& \big[\Gamma^{t}{}_{rt} + \Gamma^{r}{}_{rr} + \Gamma^{\theta}{}_{r\theta} + \Gamma^{\phi}{}_{r\phi}\big]\Gamma^{r}{}_{\phi\phi} \\
&+ \big[\cancel{\Gamma^{t}{}_{\theta t}} + \cancel{\Gamma^{r}{}_{\theta r}} + \cancel{\Gamma^{\theta}{}_{\theta\theta}} + \Gamma^{\phi}{}_{\theta\phi}\big]\Gamma^{\theta}{}_{\phi\phi} \\
=& \big[\underbrace{\frac{\nu'}{2} + \frac{\lambda'}{2}}_{=0} + \frac{1}{r} + \frac{1}{r}\big]\big[-r\cos^2\theta e^{-\lambda}\big] - \tan\theta\cos\theta\sin\theta \\
=& -2\cos^2\theta e^{-\lambda} - \sin^2\theta
\end{aligned} \tag{19}
$$

$$
\begin{aligned}
\Gamma^{\alpha}{}_{\gamma\phi}\Gamma^{\gamma}{}_{\phi\alpha} =& \cancel{\Gamma^{t}{}_{t\phi}}\Gamma^{t}{}_{\phi t} + \cancel{\Gamma^{t}{}_{r\phi}}\Gamma^{r}{}_{\phi t} + \cancel{\Gamma^{t}{}_{\theta\phi}}\Gamma^{\theta}{}_{\phi t} + \cancel{\Gamma^{t}{}_{\phi\phi}}\Gamma^{\phi}{}_{\phi t} \\
&+ \cancel{\Gamma^{r}{}_{t\phi}}\Gamma^{t}{}_{\phi r} + \cancel{\Gamma^{r}{}_{r\phi}}\Gamma^{r}{}_{\phi r} + \cancel{\Gamma^{r}{}_{\theta\phi}}\Gamma^{\theta}{}_{\phi r} + \Gamma^{r}{}_{\phi\phi}\Gamma^{\phi}{}_{\phi r} \\
&+ \cancel{\Gamma^{\theta}{}_{t\phi}}\Gamma^{t}{}_{\phi\theta} + \cancel{\Gamma^{\theta}{}_{r\phi}}\Gamma^{r}{}_{\phi\theta} + \cancel{\Gamma^{\theta}{}_{\theta\phi}}\Gamma^{\theta}{}_{\phi\theta} + \Gamma^{\theta}{}_{\phi\phi}\Gamma^{\phi}{}_{\phi\theta} \\
&+ \cancel{\Gamma^{\phi}{}_{t\phi}}\Gamma^{t}{}_{\phi\phi} + \Gamma^{\phi}{}_{r\phi}\Gamma^{r}{}_{\phi\phi} + \Gamma^{\phi}{}_{\theta\phi}\Gamma^{\theta}{}_{\phi\phi} + \cancel{\Gamma^{\phi}{}_{\phi\phi}}\Gamma^{\phi}{}_{\phi\phi} \\
=& \Gamma^{r}{}_{\phi\phi}\Gamma^{\phi}{}_{\phi r} + \Gamma^{\theta}{}_{\phi\phi}\Gamma^{\phi}{}_{\phi\theta} + \Gamma^{\phi}{}_{r\phi}\Gamma^{r}{}_{\phi\phi} + \Gamma^{\phi}{}_{\theta\phi}\Gamma^{\theta}{}_{\phi\phi} \\
=& 2\Gamma^{r}{}_{\phi\phi}\Gamma^{\phi}{}_{\phi r} + 2\Gamma^{\theta}{}_{\phi\phi}\Gamma^{\phi}{}_{\phi\theta} \\
=& -2\cancel{r}\cos^2\theta e^{-\lambda(r)}\frac{1}{\cancel{r}} - 2cos\theta\sin\theta\tan\theta \\
=& -2\cos^2\theta e^{-\lambda(r)} - 2\cancel{cos\theta}\sin\theta\frac{\sin\theta}{\cancel{cos\theta}} \\
=& -2\cos^2\theta e^{-\lambda(r)} - 2\sin^2\theta
\end{aligned} \tag{20}
$$

Combining these three equations:

$$
\begin{aligned}
R_{\phi\phi} =& \partial_\alpha\Gamma^{\alpha}{}_{\phi\phi} - \partial_\phi\Gamma^{\alpha}{}_{\phi\alpha} + \Gamma^{\alpha}{}_{\gamma\alpha}\Gamma^{\gamma}{}_{\phi\phi} - \Gamma^{\alpha}{}_{\gamma\phi}\Gamma^{\gamma}{}_{\phi\alpha} \\
=& \underbrace{-\sin^2\theta}_{\text{from (18)}} - \underbrace{\big[\cancel{2\cos^2\theta e^{-\lambda}} + \sin^2\theta\big]}_{\text{from (19)}} + \underbrace{\cancel{2\cos^2\theta e^{-\lambda(r)}} + 2\sin^2\theta}_{\text{from (20)}} \\
=& 0
\end{aligned}
$$

1.3.4 Off diagonal elements

It is easy to show that the off-diagonal terms vanish. With the exception of $R_{r\theta}$ there is no need to expand the Γ; we merely use our knowledge of which Γ and $\partial_\alpha\Gamma$ are non-zero.

$$
\begin{aligned}
R_{tr} =& \partial_\alpha\Gamma^{\alpha}{}_{tr} - \partial_r\Gamma^{\alpha}{}_{t\alpha} + \Gamma^{\alpha}{}_{\gamma\alpha}\Gamma^{\gamma}{}_{tr} - \Gamma^{\alpha}{}_{\gamma r}\Gamma^{\gamma}{}_{t\alpha} \\
=& \partial_t\Gamma^{t}{}_{tr} - \underbrace{\partial_r\Gamma^{\alpha}{}_{t\alpha}}_{=0} + \underbrace{\Gamma^{\alpha}{}_{t\alpha}}_{=0}\Gamma^{t}{}_{tr} - \underbrace{\Gamma^{t}{}_{rr}}_{=0}\Gamma^{r}{}_{tt} - \underbrace{\Gamma^{r}{}_{tr}}_{=0}\Gamma^{t}{}_{tr} \\
=& 0
\end{aligned}
$$

$$
\begin{aligned}
R_{t\theta} =& \underbrace{\partial_\alpha \Gamma^\alpha{}_{t\theta} - \partial_\theta \Gamma^\alpha{}_{t\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{t\theta}}_{\text{These terms all vanish}} - \Gamma^\alpha{}_{\gamma\theta}\Gamma^\gamma{}_{t\alpha} \\
=& -\Gamma^r{}_{\theta\theta}\underbrace{\Gamma^\theta{}_{tr}}_{=0} - \Gamma^\theta{}_{r\theta}\underbrace{\Gamma^r{}_{t\theta}}_{=0} \\
=& 0
\end{aligned}
$$

$$
\begin{aligned}
R_{t\phi} =& \underbrace{\partial_\alpha \Gamma^\alpha{}_{t\phi} - \partial_\phi \Gamma^\alpha{}_{t\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{t\phi}}_{\text{These terms all vanish}} - \Gamma^\alpha{}_{\gamma\phi}\Gamma^\gamma{}_{t\alpha} \\
=& -\Gamma^r{}_{\phi\phi}\underbrace{\Gamma^\phi{}_{tr}}_{=0} - \Gamma^\phi{}_{\theta\phi}\underbrace{\Gamma^\theta{}_{t\phi}}_{=0} - \Gamma^\phi{}_{r\phi}\underbrace{\Gamma^r{}_{t\phi}}_{=0} \\
=& 0
\end{aligned}
$$

$R_{r\theta}$ does not require a knowledge of $\mu(r)$ or $\nu(r)$.

$$
\begin{aligned}
R_{r\theta} =& \partial_\alpha \Gamma^\alpha{}_{r\theta} - \partial_\theta \Gamma^\alpha{}_{r\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{r\theta} - \Gamma^\alpha{}_{\gamma\theta}\Gamma^\gamma{}_{r\alpha} \\
=& \underbrace{\partial_r \Gamma^r{}_{r\theta}}_{=0} - \underbrace{\partial_\theta\left(\Gamma^\theta{}_{r\theta} + \Gamma^r{}_{rr}\right)}_{=0} + \Gamma^\phi{}_{\theta\phi}\Gamma^\theta{}_{r\theta} - \Gamma^r{}_{\theta\theta}\underbrace{\Gamma^\theta{}_{rr}}_{=0} - \Gamma^\theta{}_{r\theta}\underbrace{\Gamma^r{}_{r\theta}}_{=0} - \Gamma^\phi{}_{\phi\theta}\Gamma^\phi{}_{r\phi} \\
=& -\frac{1}{r^2} - \cancel{\frac{1}{r}\tan\theta} + \cancel{\frac{1}{r}\tan\theta} \\
=& 0
\end{aligned}
$$

$$
\begin{aligned}
R_{r\phi} =& \partial_\alpha \Gamma^\alpha{}_{r\phi} - \partial_\phi \Gamma^\alpha{}_{r\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{r\phi} - \Gamma^\alpha{}_{\gamma\phi}\Gamma^\gamma{}_{r\alpha} \\
=& \underbrace{\partial_\phi \Gamma^\phi{}_{r\phi}}_{=0} - \underbrace{\partial_\phi \Gamma^\alpha{}_{r\alpha}}_{=0} + \underbrace{\Gamma^\alpha{}_{\phi\alpha}}_{=0}\Gamma^\phi{}_{r\phi} - \Gamma^\phi{}_{\theta\phi}\underbrace{\Gamma^\theta{}_{r\phi}}_{=0} - \Gamma^\phi{}_{r\phi}\underbrace{\Gamma^r{}_{r\phi}}_{=0} \\
=& 0
\end{aligned}
$$

$$
\begin{aligned}
R_{\theta\phi} =& \partial_\alpha \Gamma^\alpha{}_{\theta\phi} - \partial_\phi \Gamma^\alpha{}_{\theta\alpha} + \Gamma^\alpha{}_{\gamma\alpha}\Gamma^\gamma{}_{\theta\phi} - \Gamma^\alpha{}_{\gamma\phi}\Gamma^\gamma{}_{\theta\alpha} \\
=& \underbrace{\partial_\phi \Gamma^\phi{}_{\theta\phi}}_{=0} - \underbrace{\partial_\phi \Gamma^\alpha{}_{\theta\alpha}}_{=0} + \underbrace{\Gamma^\alpha{}_{\phi\alpha}}_{=0}\Gamma^\phi{}_{\theta\phi} - \underbrace{\Gamma^\phi{}_{\phi\phi}}_{=0}\Gamma^\phi{}_{\theta\phi} \\
=& 0
\end{aligned}
$$

2 Gravitational waves

Gravitational waves are presented, in [1, Lecture 10]. In this section the coordinates are denoted: (t, x, y, x). The summation convention is observed for all other indices: e.g. ${\Gamma^\alpha}_{\gamma\alpha}$ denotes summation over the dummy index α, but ${\Gamma^x}_{\gamma x}$ is not summed.

2.1 Linearized Field Equations

From [1, Lecture 10]:

$$\begin{aligned}
{\Gamma^\tau}_{\mu\nu} =& \frac{1}{2}\big[\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}\big] g^{\sigma\tau} \\
=& \frac{1}{2}\big[\partial_\mu h_{\sigma\nu} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}\big]\big[\eta^{\sigma\tau} - O(h)\big] \\
=& \frac{1}{2}\big[\partial_\mu h_{\sigma\nu} + \partial_\nu h_{\mu\sigma} - \partial_\sigma h_{\mu\nu}\big]\eta^{\sigma\tau} - O(h^2)
\end{aligned} \tag{21}$$

2.2 Riemann and Ricci Tensors

From [1, Lecture 3]:

$$\begin{aligned}
{R^\alpha}_{\beta\gamma\delta} =& \partial_\gamma {\Gamma^\alpha}_{\beta\delta} - \partial_\delta {\Gamma^\alpha}_{\beta\gamma} + O(\Gamma^2). \text{ Now using } o(\Gamma) = o(h) \text{ from (21)} \\
=& \frac{1}{2}\big[\partial_\gamma\partial_\beta h_{\sigma\delta} + \partial_\gamma\partial_\delta h_{\beta\sigma} - \partial_\gamma\partial_\sigma h_{\beta\delta}\big]\eta^{\sigma\alpha} - \frac{1}{2}\big[\partial_\delta\partial_\gamma h_{\sigma\beta} + \partial_\delta\partial_\beta h_{\gamma\sigma} - \partial_\delta\partial_\sigma h_{\beta\gamma}\big]\eta^{\sigma\alpha} + O(h^2) \\
=& \frac{1}{2}\big[\partial_\gamma\partial_\beta h_{\sigma\delta} + \partial_\gamma\partial_\delta h_{\beta\sigma} - \partial_\gamma\partial_\sigma h_{\beta\delta} - \partial_\delta\partial_\gamma h_{\sigma\beta} - \partial_\delta\partial_\beta h_{\gamma\sigma} + \partial_\delta\partial_\sigma h_{\beta\gamma}\big]\eta^{\sigma\alpha} + O(h^2)
\end{aligned}$$

Dropping $O(h^2)$ and contracting

$$\begin{aligned}
R_{\beta\delta} =& {R^\alpha}_{\beta\alpha\delta} \\
=& \frac{1}{2}\big[\partial_\alpha\partial_\beta h_{\sigma\delta} + \cancel{\partial_\alpha\partial_\delta h_{\beta\sigma}} - \partial_\alpha\partial_\sigma h_{\beta\delta} - \cancel{\partial_\delta\partial_\alpha h_{\sigma\beta}} - \partial_\delta\partial_\beta h_{\alpha\sigma} + \partial_\delta\partial_\sigma h_{\beta\alpha}\big]\eta^{\sigma\alpha} \\
=& \frac{1}{2}\big[\partial_\alpha\partial_\beta h_{\sigma\delta} - \partial_\alpha\partial_\sigma h_{\beta\delta} - \partial_\delta\partial_\beta h_{\alpha\sigma} + \partial_\delta\partial_\sigma h_{\beta\alpha}\big]\eta^{\sigma\alpha} \\
=& 0 \text{ in empty space.}
\end{aligned} \tag{22}$$

Now $\eta^{00} = -1$, $\eta^{ii} = +1$, and all other values are zero, so Einstein's field equations become:

$$\begin{aligned}
&\sum_i \big[\underbrace{\partial_i\partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i\partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta\partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta\partial_i h_{\beta i}}_{\text{(d)}}\big] \\
&= \underbrace{\partial_0\partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0\partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta\partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta\partial_0 h_{\beta 0}}_{\text{(h)}}
\end{aligned} \tag{23}$$

2.3 Ansatz

We will look for plane wave solutions propagating along the z axis.

$$\begin{aligned}
h_{\mu\nu}(x,t) =& h^0_{\mu\nu} \sin\big(k(t-z)\big) \\
\partial_0^2 h_{\mu\nu} =& -k^2 h_{\mu\nu} \\
\partial_z^2 h_{\mu\nu} =& -k^2 h_{\mu\nu} \\
\partial_0\partial_z h_{\mu\nu} =& k^2 h_{\mu\nu} \\
\partial_z\partial_0 h_{\mu\nu} =& k^2 h_{\mu\nu}
\end{aligned}$$

All other partial derivatives are zero.

The terms on the left hand side of (23) become:

$$\begin{aligned}
(a) \sum_i \partial_i\partial_\beta h_{i\delta} =& \begin{cases} \partial_z\partial_\beta h_{z\delta},\ \beta \in \{z,t\} \\ 0, \text{ otherwise.} \end{cases} \\
(b) \sum_i \partial_i\partial_i h_{\beta\delta} =& \partial_z\partial_z h_{\beta\delta} \\
(c) \sum_i \partial_\delta\partial_\beta h_{ii} =& \begin{cases} \partial_\delta\partial_\beta \sum_i h_{ii},\ \beta\&\delta \in \{z,t\} \\ 0, \text{ otherwise.} \end{cases} \\
(d) \sum_i \partial_\delta\partial_i h_{\beta i} =& \begin{cases} \partial_\delta\partial_z h_{\beta z},\ \delta \in \{z,t\} \\ 0, \text{ otherwise} \end{cases}
\end{aligned}$$

The terms on the right hand side of (23) become:

$$\begin{aligned}
&(e)\ \partial_0\partial_\beta h_{0\delta} = 0 \text{ unless } \beta \in \{z,t\} \\
&(f)\ \partial_0\partial_0 h_{\beta\delta} \neq 0 \\
&(g)\ \partial_\delta\partial_\beta h_{00} = 0 \text{ unless } \beta\&\delta \in \{z,t\} \\
&(h)\ \partial_\delta\partial_0 h_{\beta 0} = 0 \text{ unless } \delta \in \{z,t\}
\end{aligned}$$

2.4 Detailed Constraints

Substituting $\beta = 0, \delta = 0$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i \partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i \partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta \partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta \partial_i h_{\beta i}}_{\text{(d)}} \big] \\
&\quad = \underbrace{\partial_0 \partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta \partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta \partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{\partial_z \partial_t h_{zt}}_{\text{(a)}} - \underbrace{\partial_z \partial_z h_{tt}}_{\text{(b)}} - \underbrace{\partial_t \partial_t \sum_i h_{ii}}_{\text{(c)}} + \underbrace{\partial_t \partial_z h_{tt}}_{\text{(d)}} \\
&\quad = \underbrace{\cancel{\partial_t \partial_t h_{tt}}}_{\text{(e)}} - \underbrace{\cancel{\partial_t \partial_t h_{tt}}}_{\text{(f)}} - \underbrace{\cancel{\partial_t \partial_t h_{tt}}}_{\text{(g)}} + \underbrace{\cancel{\partial_t \partial_t h_{tt}}}_{\text{(h)}} \\
0 =& \partial_z \partial_t h_{zt} - \partial_z \partial_z h_{tt} - \partial_t \partial_t \sum_i h_{ii} + \partial_t \partial_z h_{tz} \\
=& k^2 \big[h^0_{zt} + h^0_{tt} + \sum_i h^0_{ii} + h^0_{tz} \big] \sin\big(k(t-z)\big) \\
0 =& 2h^0_{zt} + h^0_{tt} + \sum_i h^0_{ii}
\end{aligned}
\tag{24}
$$

Substituting $\beta = 0, \delta = 1$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i \partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i \partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta \partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta \partial_i h_{\beta i}}_{\text{(d)}} \big] \\
&\quad = \underbrace{\partial_0 \partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta \partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta \partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \partial_z \partial_t h_{zx} - \partial_z \partial_z h_{tx} = \cancel{\partial_t \partial_t h_{tx}} - \cancel{\partial_t \partial_t h_{tx}} \\
0 =& k^2 \big[h^0_{zx} + h^0_{tx} \big] \sin\big(k(t-z)\big) \\
0 =& h^0_{zx} + h^0_{tx}
\end{aligned}
\tag{25}
$$

Substituting $\beta = 0, \delta = 2$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i \partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i \partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta \partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta \partial_i h_{\beta i}}_{\text{(d)}} \big] \\
&\quad = \underbrace{\partial_0 \partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta \partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta \partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{\partial_z \partial_t h_{zy}}_{\text{(a)}} - \underbrace{\partial_z \partial_z h_{ty}}_{\text{(b)}} = \underbrace{\cancel{\partial_t \partial_t h_{ty}}}_{\text{(e)}} - \underbrace{\cancel{\partial_t \partial_t h_{ty}}}_{\text{(f)}} \\
0 =& k^2 \big[h^0_{zy} + h^0_{ty} \big] \sin\big(k(t-z)\big) \\
0 =& h^0_{zy} + h^0_{ty}
\end{aligned}
\tag{26}
$$

Substituting $\beta = 0, \delta = 3$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i\partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i\partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta\partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta\partial_i h_{\beta i}}_{\text{(d)}}\big] \\
&\quad = \underbrace{\partial_0\partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0\partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta\partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta\partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{\partial_z\partial_t h_{zz}}_{\text{(a)}} - \underbrace{\cancel{\partial_z\partial_z h_{tz}}}_{\text{(b)}} - \underbrace{\partial_z\partial_t \sum_i h_{ii}}_{\text{(c)}} + \underbrace{\cancel{\partial_z\partial_z h_{tz}}}_{\text{(d)}} \\
&\quad = \underbrace{\cancel{\partial_t\partial_t h_{tz}}}_{\text{(e)}} - \underbrace{\cancel{\partial_t\partial_t h_{tz}}}_{\text{(f)}} - \underbrace{\cancel{\partial_z\partial_t h_{tt}}}_{\text{(g)}} + \underbrace{\cancel{\partial_z\partial_t h_{tt}}}_{\text{(h)}} \\
&0 = k^2\left[h^0_{xx} + h^0_{yy}\right]\sin\big(k(t-z)\big) \\
&0 = h^0_{xx} + h^0_{yy}
\end{aligned}
\tag{27}
$$

Substituting $\beta = 1, \delta = 1$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i\partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i\partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta\partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta\partial_i h_{\beta i}}_{\text{(d)}}\big] \\
&\quad = \underbrace{\partial_0\partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0\partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta\partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta\partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{0}_{\text{(a)}} - \underbrace{\partial_z\partial_z h_{\beta\delta}}_{\text{(b)}} - \underbrace{0}_{\text{(c)}} + \underbrace{0}_{\text{(d)}} \\
&\quad = \underbrace{0}_{\text{(e)}} - \underbrace{\partial_0\partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{0}_{\text{(g)}} + \underbrace{0}_{\text{(h)}} \\
&0 = \partial_z\partial_z h_{xx} - \partial_0\partial_0 h_{xx} \\
&0 = k^2\left[h^0_{xx} - h^0_{xx}\right]\sin\big(k(t-z)\big) \text{ an empty constraint}
\end{aligned}
$$

Substituting $\beta = 1, \delta = 2$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i \partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i \partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta \partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta \partial_i h_{\beta i}}_{\text{(d)}} \big] \\
&\quad = \underbrace{\partial_0 \partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta \partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta \partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{0}_{\text{(a)}} - \underbrace{\partial_z \partial_z h_{xy}}_{\text{(b)}} - \underbrace{0}_{\text{(c)}} + \underbrace{0}_{\text{(d)}} \\
&\quad = \underbrace{0}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{xy}}_{\text{(f)}} - \underbrace{0}_{\text{(g)}} + \underbrace{0}_{\text{(h)}} \\
&0 = \partial_z \partial_z h_{\beta\delta} - \partial_0 \partial_0 h_{\beta\delta} \\
&0 = k^2 \big[h^0_{xy} - h^0_{xy} \big] \sin\big(k(t-z)\big) \text{ an empty constraint}
\end{aligned}
$$

Substituting $\beta = 1, \delta = 3$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i \partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i \partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta \partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta \partial_i h_{\beta i}}_{\text{(d)}} \big] \\
&\quad = \underbrace{\partial_0 \partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta \partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta \partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{0}_{\text{(a)}} - \underbrace{\cancel{\partial_z \partial_z h_{xz}}}_{\text{(b)}} - \underbrace{0}_{\text{(c)}} + \underbrace{\cancel{\partial_z \partial_z h_{xz}}}_{\text{(d)}} \\
&\quad = \underbrace{0}_{\text{(e)}} - \underbrace{\partial_t \partial_t h_{xz}}_{\text{(f)}} - \underbrace{0}_{\text{(g)}} + \underbrace{\partial_z \partial_t h_{xt}}_{\text{(h)}} \\
&0 = \partial_t \partial_t h_{xz} - \partial_z \partial_t h_{xt} \\
&\quad = k^2 \big[h^0_{xz} + h^0_{xt} \big] \sin\big(k(t-z)\big) \\
&0 = h^0_{xz} + h^0_{xt}, \text{ which is the same as (25)}
\end{aligned}
$$

Substituting $\beta = 2, \delta = 2$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i\partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i\partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta\partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta\partial_i h_{\beta i}}_{\text{(d)}}\big] \\
&\quad = \underbrace{\partial_0\partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0\partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta\partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta\partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{0}_{\text{(a)}} - \underbrace{\partial_z\partial_z h_{yy}}_{\text{(b)}} - \underbrace{0}_{\text{(c)}} + \underbrace{0}_{\text{(d)}} \\
&\quad = \underbrace{0}_{\text{(e)}} - \underbrace{\partial_t\partial_t h_{yy}}_{\text{(f)}} - \underbrace{0}_{\text{(g)}} + \underbrace{0}_{\text{(h)}} \\
0 =& \partial_z\partial_z h_{yy} - \partial_t\partial_t h_{yy} \\
=& k^2\left[h^0_{yy} - h^0_{yy}\right]\sin\big(k(t-z)\big) \\
0 =& h^0_{yy} - h^0_{yy} \text{ an empty constraint}
\end{aligned}
$$

Substituting $\beta = 2, \delta = 3$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i\partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i\partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta\partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta\partial_i h_{\beta i}}_{\text{(d)}}\big] \\
&\quad = \underbrace{\partial_0\partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0\partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta\partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta\partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{0}_{\text{(a)}} - \underbrace{\cancel{\partial_z\partial_z h_{yz}}}_{\text{(b)}} - \underbrace{0}_{\text{(c)}} + \underbrace{\cancel{\partial_z\partial_z h_{yz}}}_{\text{(d)}} \\
&\quad = \underbrace{0}_{\text{(e)}} - \underbrace{\partial_t\partial_t h_{yz}}_{\text{(f)}} - \underbrace{0}_{\text{(g)}} + \underbrace{\partial_z\partial_t h_{yt}}_{\text{(h)}} \\
0 =& \partial_t\partial_t h_{yz} - \partial_z\partial_t h_{yt} \\
=& k^2\left[h_{yz} + h_{yt}\right]\sin\big(k(t-z)\big) \\
0 =& h_{yz} + h_{yt}, \text{ which is the same as (26)}
\end{aligned}
$$

Substituting $\beta = 3, \delta = 3$ in (23):

$$
\begin{aligned}
&\sum_i \big[\underbrace{\partial_i \partial_\beta h_{i\delta}}_{\text{(a)}} - \underbrace{\partial_i \partial_i h_{\beta\delta}}_{\text{(b)}} - \underbrace{\partial_\delta \partial_\beta h_{ii}}_{\text{(c)}} + \underbrace{\partial_\delta \partial_i h_{\beta i}}_{\text{(d)}} \big] \\
&\quad = \underbrace{\partial_0 \partial_\beta h_{0\delta}}_{\text{(e)}} - \underbrace{\partial_0 \partial_0 h_{\beta\delta}}_{\text{(f)}} - \underbrace{\partial_\delta \partial_\beta h_{00}}_{\text{(g)}} + \underbrace{\partial_\delta \partial_0 h_{\beta 0}}_{\text{(h)}} \\
&\quad \underbrace{\cancel{\partial_z \partial_z h_{zz}}}_{\text{(a)}} - \underbrace{\cancel{\partial_z \partial_z h_{zz}}}_{\text{(b)}} - \underbrace{\partial_z \partial_z \sum_i h_{ii}}_{\text{(c)}} + \underbrace{\partial_z \partial_z h_{zz}}_{\text{(d)}} \\
&\quad = \underbrace{\partial_t \partial_z h_{tz}}_{\text{(e)}} - \underbrace{\partial_t \partial_t h_{zz}}_{\text{(f)}} - \underbrace{\partial_z \partial_z h_{tt}}_{\text{(g)}} + \underbrace{\partial_z \partial_t h_{zt}}_{\text{(h)}} \\
0 =& \partial_z \partial_z \big(h_{xx} + h_{yy} \big) + \partial_t \partial_z h_{tz} - \partial_t \partial_t h_{zz} - \partial_z \partial_z h_{tt} + \partial_z \partial_t h_{zt} \\
=& k^2 \big[- h^0_{xx} - h^0_{yy} + h^0_{tz} + h^0_{zz} + h^0_{tt} + h^0_{zt} \big] \sin\big(k(t-z)\big) l_\delta \partial_\beta h_{00} - \partial_\delta \partial_0 h_{\beta 0} \\
0 =& \underbrace{-h^0_{xx} - h^0_{yy}}_{= 0 \text{ from } (27)} + h^0_{tz} + h^0_{zz} + h^0_{tt} + h^0_{zt} \\
=& + h^0_{zz} + h^0_{tt} + 2h^0_{zt} \qquad (28)
\end{aligned}
$$

2.5 Summary of Constraints

Table 1 shows there are only 5 constraints on the 10 parameters $h^0_{\mu\nu}$, so there are indeed plane wave solutions.

Table 1: Constraints on the parameters $h^0_{\mu\nu}$

#	Constraint
(24)	$2h^0_{zt} + h^0_{tt} + \underbrace{h^0_{xx} + h^0_{yy}}_{= 0 \text{ from } (27)} + h^0_{zz} = 0$
(25)	$h^0_{zx} + h^0_{tx} = 0$
(26)	$h^0_{zy} + h^0_{ty} = 0$
(27)	$h^0_{xx} + h^0_{yy} = 0$
(28)	$h^0_{zz} + h^0_{tt} + 2h^0_{zt} = 0$

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